Ahmet Emir Şener

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Education

Boğaziçi University — İstanbul, Turkey
M.S. in Industrial Engineering (GPA: 3.94/4.00) — January 2025 – Present

- **Thesis:** New Convex Relaxations for Global Optimization Problems in Water Networks
- **Advisors:** Necati Aras and Burak Kocuk

B.S. in Industrial Engineering (GPA: 3.64/4.00) — September 2021 – January 2025

- **Thesis:** Assessment of Alternative Bidding Strategies in Multi-Actor Context of the Turkish Electricity Market
- **Advisors:** Gönenç Yücel and Mustafa Gökçe Baydoğan

Üsküdar American Academy — İstanbul, Turkey
High School Diploma — September 2016 – June 2021

Academic Appointments

Boğaziçi University — İstanbul, Turkey
Department of Industrial Engineering
Graduate Assistant — October 2026 – Present

Manuscripts

1. **Ahmet Emir Şener**, Burak Kocuk, and Tuğçe Yüksel (2026). Benders Decomposition with Partial Non-Anticipativity Relaxation for Multi-Stage Stochastic Clean Energy Transition Planning. Submitted to *European Journal of Operational Research*. arXiv:2608.14105.
2. **Ahmet Emir Şener**, Burak Kocuk, and Tuğçe Yüksel (2025). A Dynamic Strategic Plan for Transition to Campus-Scale Clean Electricity Using Multi-Stage Stochastic Programming. Submitted to *The Energy Journal*. arXiv:2506.16423.

Refereed Conference Proceedings

1. **Ahmet Emir Şener** and Gönenç Yücel (2025). A Dynamic Model of Workforce Management During Seasonal Epidemics. In *Proceedings of the International System Dynamics Conference*, Boston, USA.

Funded Research Experience

Sabancı University — İstanbul, Turkey
Research Scholar, TÜBİTAK Program 1001

New Convex Relaxations for Global Optimization Problems in Water Networks — February 2026 – Present

- Derive convex hull descriptions of hydraulic relations and propose second-order cone outer approximation methods for power cones. Implement cutting-plane algorithms to solve nonconvex mixed-integer nonlinear programs arising in water distribution network design and operation.

Strategic Planning for Transition to Next Generation Clean Energy Technologies using Multi-Stage Stochastic Programming — February 2024 – February 2026

- Developed optimization models and solution methodologies for large-scale multi-stage stochastic mixed-integer programs to support multi-vector clean energy transition planning under technological and operational uncertainty.

Conference and Workshop Presentations

1. New Convex Relaxations for Global Optimization Problems in Water Networks. Workshop on Graph Theory and Its Applications XIII, Boğaziçi University, İstanbul, Turkey, October 2026.
2. Benders Decomposition with Partial Non-Anticipativity Relaxation for Multi-Stage Stochastic Clean Energy Transition Planning. YAEM 2026, İstanbul Technical University, İstanbul, Turkey, July 2026.
3. A Dynamic Strategic Plan for Transition to Campus-Scale Clean Electricity Using Multi-Stage Stochastic Programming. YAEM 2025, Gazi University, Ankara, Turkey, June 2025.
4. Clean Electricity Transition under Technology Cost and Efficiency Uncertainty for Distributed Energy Systems. YAEM 2024, Karadeniz Technical University, Trabzon, Turkey, October 2024.

Teaching Experience

Boğaziçi University İstanbul, Turkey
Department of Industrial Engineering

- Graduate Teaching Assistant — September 2025 – January 2026
  IE 550 – Dynamics of Socio-Economic Systems
- Undergraduate Teaching Assistant — September 2024 – January 2025
  IE 306 – Systems Simulation

Awards and Honors

High Honor Degree, Boğaziçi University — January 2025
University Entrance Examination, quantitative track: 929th nationwide — July 2021
Sports Scholarship, Üsküdar American Academy — September 2018 – June 2021

Industry Experience

Doğa Ambalaj ve Etiket August – September 2024
Production Planning Intern

- Developed mathematical optimization models to schedule label printing jobs, accounting for order due dates, priority constraints, and sequence-dependent color changeover times.

Icron Technologies August – September 2023
Research and Development Intern

- Integrated the OpenAI API's function-calling feature into ICRON's software, enabling users to ask natural-language questions about results from mathematical optimization models built within the platform.

Birleşim Mühendislik June – September 2022
Engineering Intern

- Formulated exact and approximate functions for calculating sheet metal surface areas from steel air duct specifications and integrated these calculations into an ERP system using the AutoCAD API.

Skills

Computing: Python, R, Gurobi, CPLEX, MOSEK, Ipopt, BARON, SIMAN, AnyLogic, Vensim, LaTeX
Languages: Turkish, English